



SUJET STATAPP 338 - 2026-2027



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Organisme : Société Générale

Catégorie : Finance

Sujet :

Explaining Cheapest-to-Deliver (CTD) Bond Transitions in Government Bond Futures

Description du sujet :

The Cheapest-to-Deliver (CTD) bond plays a central role in the pricing and risk management of bond futures. Although a futures contract is linked to a basket of eligible bonds, the short position will normally deliver the bond that minimizes delivery costs. The identity of this bond may change over time as interest rates, yield curve shapes, bond spreads, and financing conditions evolve.

The objective of this project is to forecast CTD switches and identify the key market factors driving changes in the cheapest-to-deliver bond. Using historical data for a government bond futures contract, the students will investigate which market variables explain transitions from one CTD bond to another and assess the impact of this change on a short bond futures position.

Données du sujet :

Bond Futures market data

Modèles envisagés du sujet :

regression, random forest, XGboost and HW for the stochastic model

Bibliographie du sujet :

<https://www.math.vu.nl/~sbhulai/papers/thesis-vanstraaten.pdf>

https://www.m-x.ca/f_publications_en/futures_flash_article5_en.pdf

Notation du sujet :

3

Commentaires :

Ok même si je ne suis pas expert du sujet.